ASSIGNMENT 4

Spoken Arabic Digit Sequential Classification

Task 2 | Pre-computed MFCC Sequences | Deep Learning

Best model	BiGRU-L2-H128+Att
Final training accuracy	100.00%
Validation accuracy	99.33%
Held-out test accuracy	97.95% (2,200 utterances)
Configurations evaluated	17 (12 main-grid + 5 bonus)

Abstract

This experiment classifies spoken Arabic digits from numerical sequences of Mel-Frequency Cepstral Coefficients (MFCCs). Each utterance is represented as a variable-length sequence of 13 acoustic features. Vanilla RNN, LSTM and GRU models were compared across different hidden sizes and depths, followed by bidirectional and temporal-attention experiments. Model selection used validation accuracy, while the official test split was reserved for final evaluation.

Main finding

The two-layer bidirectional GRU with temporal attention achieved the strongest test result in the notebook: 97.95%. Gated recurrent models substantially outperformed the vanilla RNN on this speech task, and bidirectional processing improved recognition of utterances from unseen speakers.

1. Dataset and sequential representation

The Spoken Arabic Digit dataset contains utterances for ten classes (digits 0-9). The supplied text files already contain MFCC feature frames, so no raw-audio decoding or MFCC extraction was required. Each frame contains 13 coefficients, and the number of frames varies by utterance.

Dataset split and preprocessing

Subset	Utterances	Class balance	Sequence length in subset
Training	6,000	600 per digit	4-93 frames
Validation	600	60 per digit	14-62 frames
Test	2,200	220 per digit	7-83 frames
All loaded utterances	8,800	10 balanced classes	4-93 frames; mean 39.8

The training text file contains 6,600 utterance blocks. For each digit, the final 60 blocks (corresponding to the last six speakers under the documented file ordering) were set aside for validation, leaving 600 training utterances per digit. The separate test file contains 2,200 utterances from 22 other speakers. MFCC coefficients were standardised using statistics calculated from the training split only.

Variable-length batches were zero-padded and the true sequence lengths were retained. Padding was excluded from the decision: bidirectional models used packed sequences, while attention scores were masked at padded frames.

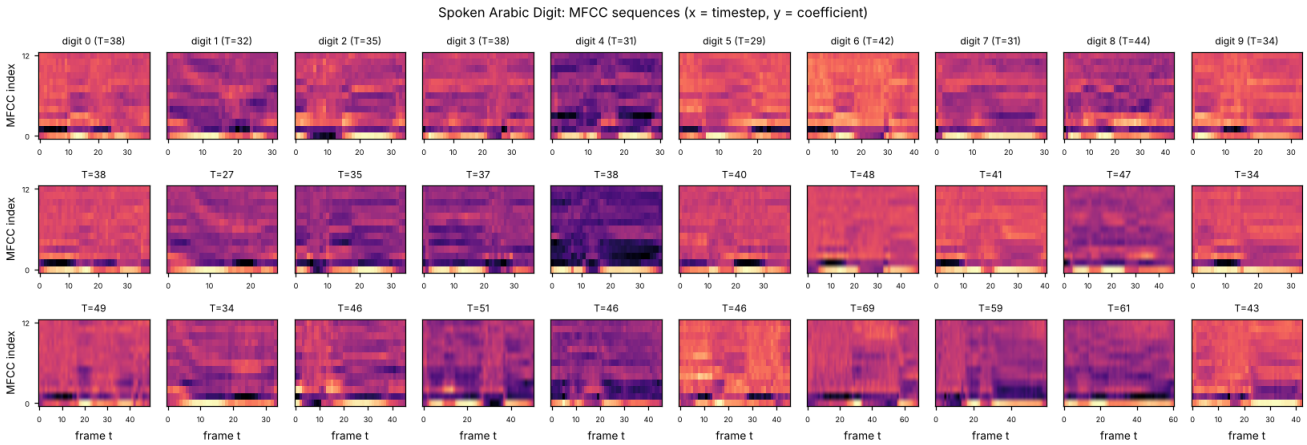


Figure 1. MFCC heatmaps for three training utterances per digit. Horizontal position is the frame index; vertical position is the MFCC coefficient.



Figure 2. Distribution of utterance lengths across the loaded dataset.

2. Models and training procedure

A shared sequence-classification pipeline was used for the experiments. A recurrent encoder processes the MFCC frames and a linear classification layer maps the representation to ten digit classes.

Architectures explored

Main grid (12 configurations): Vanilla RNN, LSTM and GRU, each with one or two recurrent layers and 64 or 128 hidden units. **Bonus grid (5 configurations):** one-layer bidirectional LSTM, one-layer bidirectional GRU, LSTM with temporal attention, GRU with temporal attention, and a two-layer bidirectional GRU with attention.

For ordinary unidirectional models, the representation is the hidden state at the utterance's true final frame. Attention models compute a weighted sum of valid timestep representations. The best overall architecture combines two bidirectional GRU layers with temporal attention.

Training settings

Setting	Value
Optimiser	Adam
Initial learning rate	0.001; multiplied by 0.85 after each epoch
Batch size (Task 2)	64
Gradient clipping	Maximum gradient norm 5.0
Convergence rule	Stop when |L(k) - L(k-1)| < 1e-4
Maximum epoch cap	60 (safety cap; not reached by any run)
Random seed	42

The stopping condition follows the assignment requirement: the absolute difference between successive epochs' average training cross-entropy must fall below 10^-4. The notebook also decays the learning rate each epoch, which helps the loss change settle below this threshold. All 17 configurations converged before the safety cap.

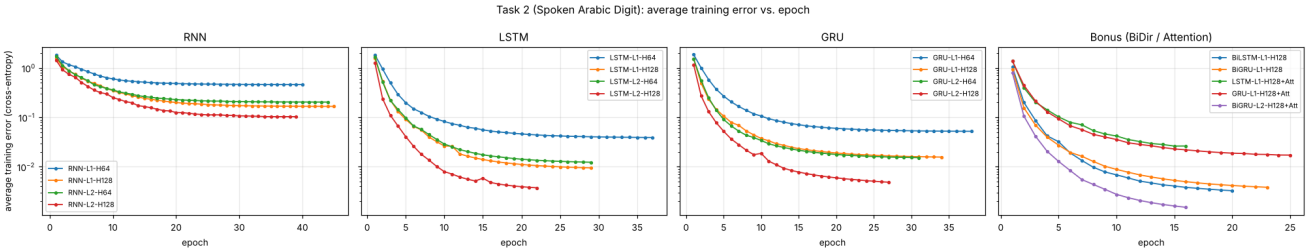


Figure 3. Average training loss versus epoch for all Task 2 hyperparameter configurations. Each curve ends at the epoch where the convergence condition was met. The vertical scale is logarithmic.

3. Hyperparameter results

All configurations below are reproduced from the notebook summary table. Accuracy values are percentages. “BiDir” indicates bidirectional processing; “Attn” indicates temporal attention pooling. The best model is selected by validation accuracy, not by test accuracy.

Model	Params	Epochs	Final loss	Train %	Val %	Test %
RNN-L1-H64	5,706	40	0.46273	82.55	83.83	76.41
RNN-L1-H128	19,594	45	0.16699	95.72	95.67	88.86
RNN-L2-H64	14,026	44	0.20439	94.58	91.00	85.59
RNN-L2-H128	52,618	39	0.10349	97.33	92.83	90.18
LSTM-L1-H64	20,874	37	0.03924	99.28	97.50	92.86
LSTM-L1-H128	74,506	29	0.00950	99.88	98.00	94.27
LSTM-L2-H64	54,154	29	0.01237	99.92	98.17	94.64
LSTM-L2-H128	206,602	22	0.00376	99.95	98.00	94.50
GRU-L1-H64	15,818	38	0.05229	98.98	99.00	94.05
GRU-L1-H128	56,202	34	0.01573	99.75	98.83	94.82
GRU-L2-H64	40,778	31	0.01532	99.78	98.50	93.86
GRU-L2-H128	155,274	27	0.00488	99.95	98.83	95.27
BiLSTM-L1-H128	149,002	20	0.00328	99.98	95.83	96.95
BiGRU-L1-H128	112,394	23	0.00385	99.97	98.50	96.73
LSTM-L1-H128+Att	91,146	16	0.02637	99.47	96.33	95.32
GRU-L1-H128+Att	72,842	25	0.01726	99.68	97.33	96.68
BiGRU-L2-H128+Att	**474,890**	**16**	**0.00151**	**100.00**	**99.33**	**97.95**

Interpretation: the best plain RNN reached 88.86% test accuracy, the best plain LSTM reached 94.64%, and the best plain GRU reached 94.05%. The best result overall was 97.95%, obtained by BiGRU-L2-H128+Att.

4. Accuracy comparison and selected model

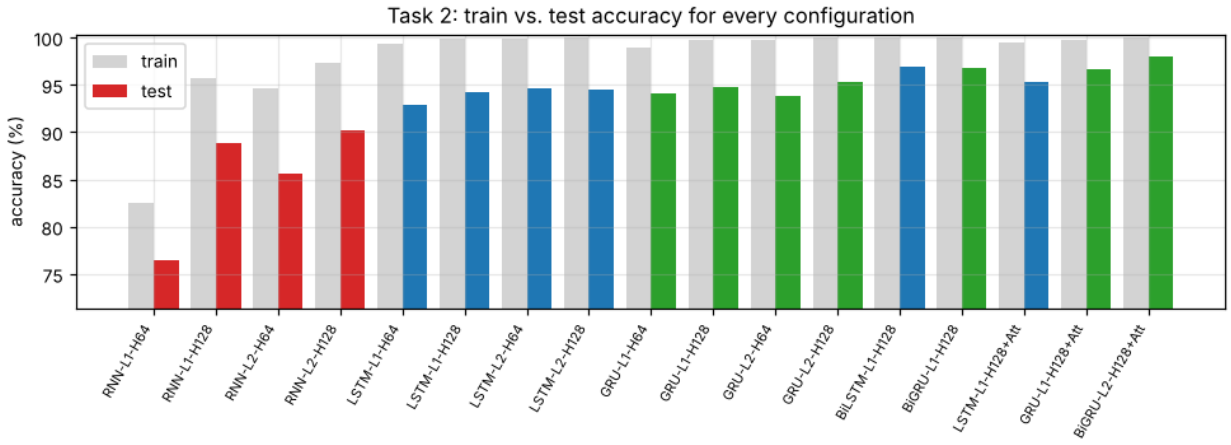


Figure 4. Training and test accuracy for all 17 configurations. The bars preserve the values plotted in the executed notebook.

Training convergence plot

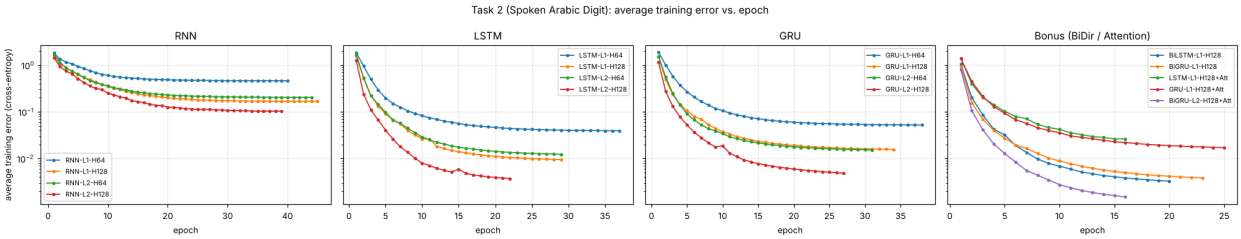


Figure 5. Average training loss by epoch for every Task 2 configuration.

Best configuration: BiGRU-L2-H128+Att

Metric	Notebook result
Architecture	2-layer bidirectional GRU, 128 units, temporal attention
Trainable parameters	474,890
Converged after	16 epochs
Final training loss	0.00151
Training accuracy	100.00%
Validation accuracy	99.33%
Test accuracy	97.95% (2,155 of 2,200 correct)

The selected model achieved the highest validation accuracy among the tested configurations. Its test performance is also the strongest, with 2,155 correctly classified utterances out of 2,200. This is a speaker-held-out evaluation, so the result reflects performance on utterances from speakers not included in training.

Per-class scores for the selected model

Digit	Precision	Recall	F1-score	Support
0	0.9459	0.9545	0.9502	220
1	0.9955	1.0000	0.9977	220
2	0.9955	0.9955	0.9955	220
3	0.9636	0.9636	0.9636	220
4	0.9690	0.9955	0.9821	220
5	0.9954	0.9909	0.9932	220
6	0.9952	0.9409	0.9673	220

Digit	Precision	Recall	F1-score	Support
7	0.9769	0.9591	0.9679	220
8	0.9865	0.9955	0.9910	220
9	0.9735	1.0000	0.9865	220
Macro / weighted avg	0.9797	0.9795	0.9795	2,200

5. Confusion matrices and error analysis

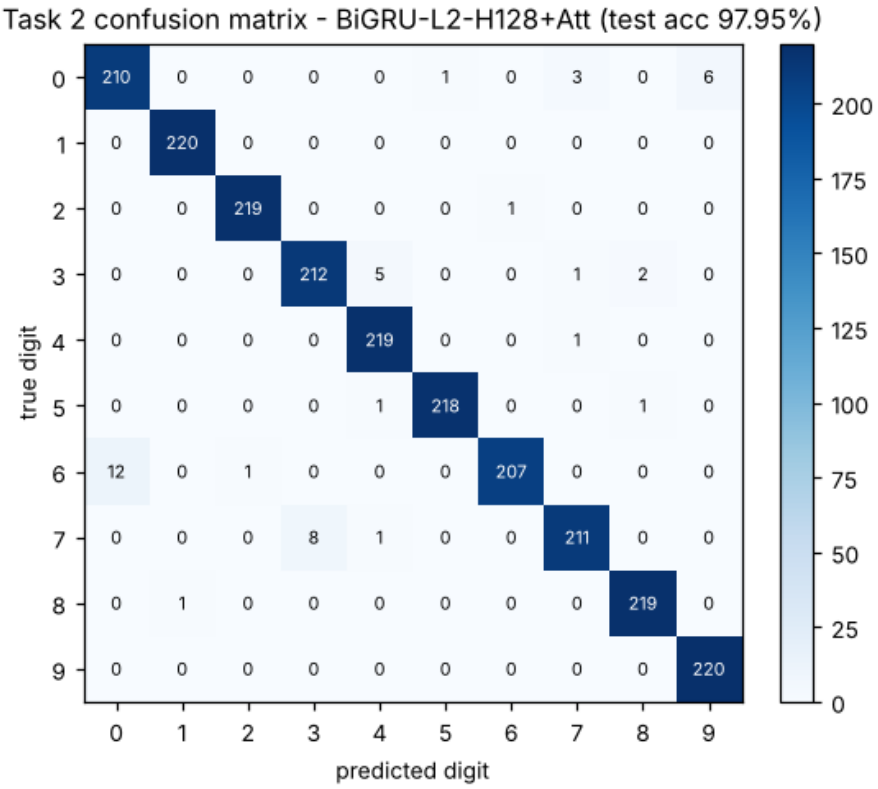


Figure 6. Confusion matrix for the selected BiGRU-L2-H128+Att model (test accuracy 97.95%). Rows represent true digits and columns represent predicted digits.

The selected model recognises digits 1 and 9 with no errors. The largest confusions are digit 6 predicted as 0 (12 utterances), digit 7 predicted as 3 (8), and digit 0 predicted as 9 (6). Digit 6 has the lowest recall (94.09%).

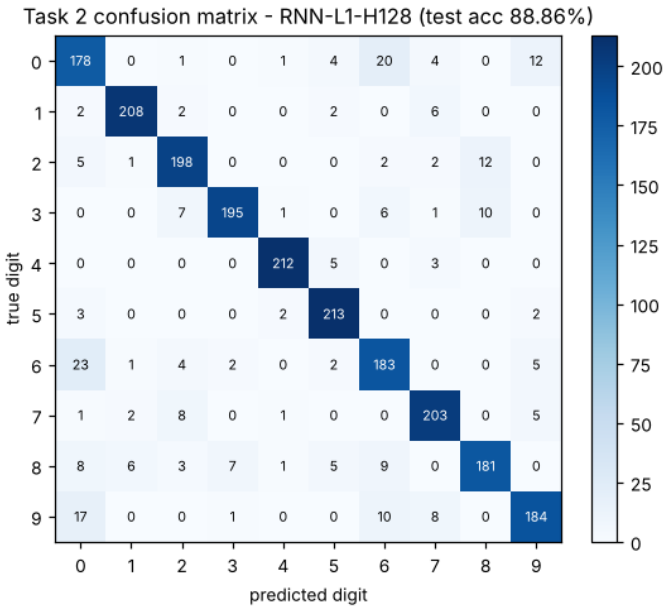


Figure 7. Confusion matrix for the best plain RNN, RNN-L1-H128 (test accuracy 88.86%).

5. Confusion matrices and error analysis (continued)

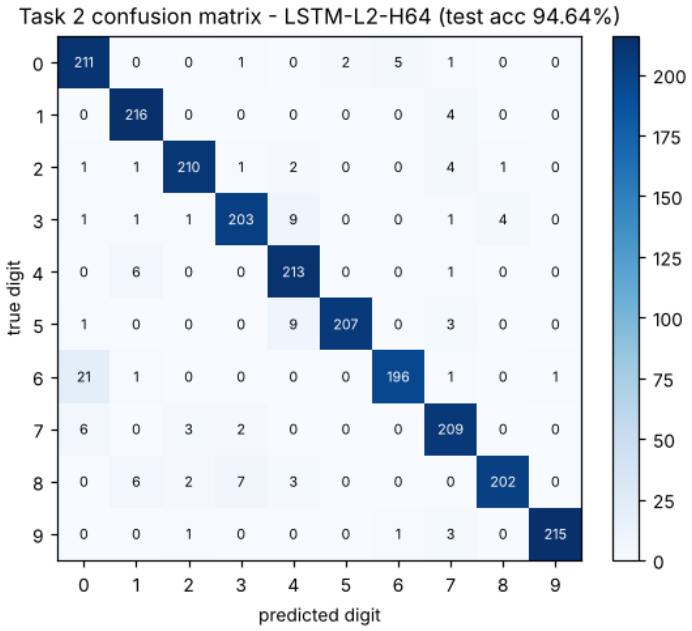


Figure 8. Confusion matrix for the best plain LSTM, LSTM-L2-H64 (test accuracy 94.64%).

The baseline matrices show that the vanilla RNN makes more errors across several classes, while the LSTM predictions are more concentrated along the diagonal. The best model further reduces errors, supporting the benefit of gated recurrence and the bidirectional-attention combination.

6. Gradient-flow analysis

The notebook evaluates gradient propagation for one-layer, 128-unit RNN, LSTM and GRU models on Task 2. The first two plots show input-gradient magnitude across long test utterances before and after training; the third tracks mean parameter-gradient norm over training epochs. These are the original notebook plots, separated here for readability.

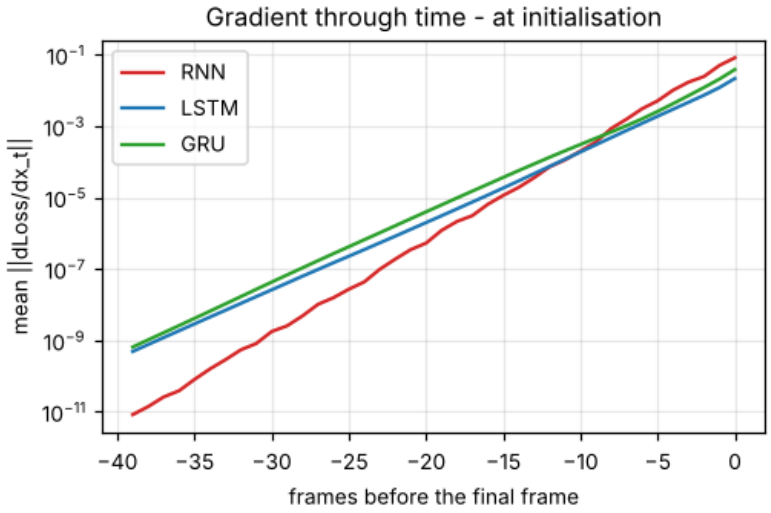


Figure 9. Input-gradient magnitude at initialisation, aligned to the final frame (0).

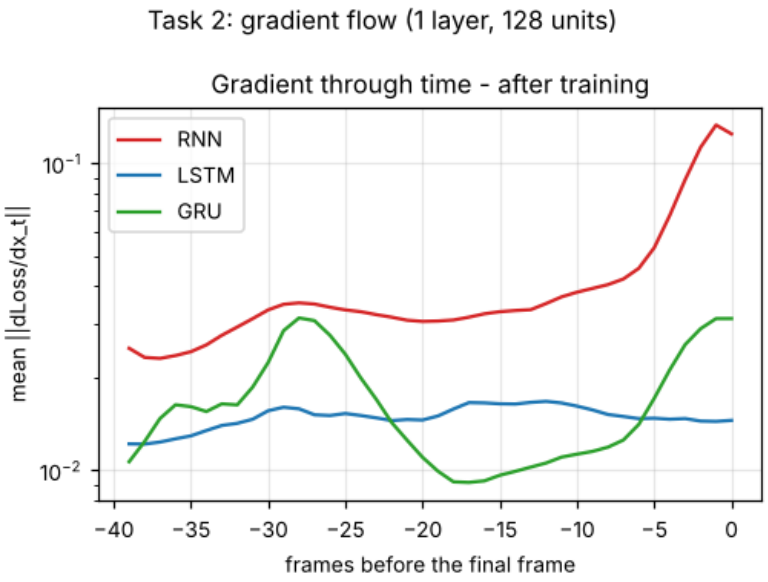


Figure 10. Input-gradient magnitude after training, aligned to the final frame (0).

6. Gradient-flow analysis (continued)

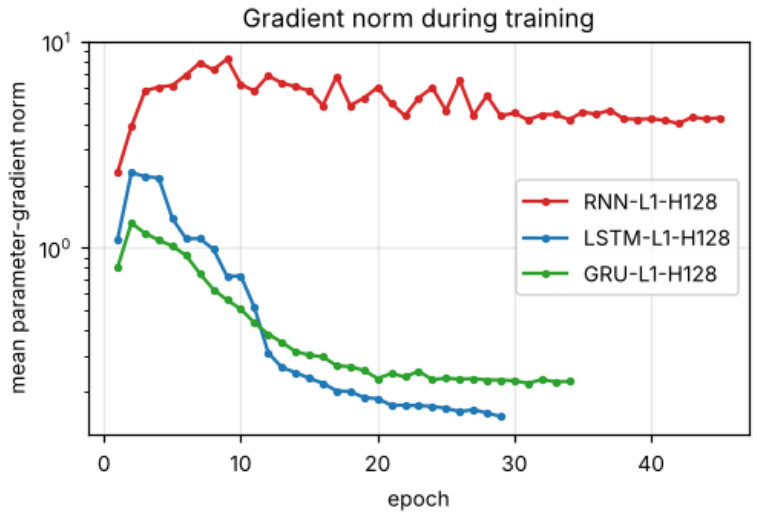


Figure 11. Mean parameter-gradient norm during training, before gradient clipping.

Before training, gradients decrease sharply with temporal distance for all three recurrent cells, with the vanilla RNN showing the strongest decay. After training, the LSTM curve is comparatively even across the 40-frame window, while the RNN is concentrated near the final frames and the GRU varies across time. During training, the RNN also exhibits larger and noisier parameter-gradient norms; the gated models decay more smoothly.

RNN, LSTM and GRU performance

Cell type	Mean train %	Mean test %	Best test %	Worst test %	Mean epochs
Vanilla RNN	92.54	85.26	90.18	76.41	42.0
LSTM	99.76	94.07	94.64	92.86	29.2
GRU	99.62	94.50	95.27	93.86	32.5

Performance. The gated LSTM and GRU models outperform the vanilla RNN on this variable-length speech task. Across the four plain configurations of each cell type, mean test accuracy is 85.26% for RNN, 94.07% for LSTM and 94.50% for GRU. The best plain GRU achieves 95.27%.

Bidirectionality and attention. The bidirectional LSTM and GRU reach 96.95% and 96.73% test accuracy respectively. Temporal attention improves the one-layer GRU result to 96.68%. Combining bidirectional processing, two GRU layers and attention gives the highest result, 97.95%. The combined model is also the largest configuration (474,890 parameters), so capacity is a possible contributor to the improvement.

Gradient flow. Before training, gradients decay strongly over long sequences for all three cells. At 40 frames, the earliest-to-latest gradient ratios reported by the notebook are 1.014e-10 for RNN, 2.262e-08 for LSTM and 1.691e-08 for GRU. After training, the corresponding ratios are 0.2002, 0.8384 and 0.3417. This suggests that trained gated models, especially the LSTM in this measurement, preserve gradient signal across the tested time span more evenly than the vanilla RNN.

Conclusion. The experiments show that recurrent gating is important for robust classification of speech sequences. Bidirectional context and attention provide further gains, and BiGRU-L2-H128+Att is the strongest of the tested models on the held-out speaker test set, at 97.95% accuracy. These results are from one random seed and one run per configuration; repeating the strongest models over multiple seeds would help establish how stable the ranking is.

Reproducibility note: This report follows the supplied notebook's recorded outputs. The training loss plots, MFCC heatmaps, accuracy chart and confusion matrices are notebook-rendered figures rather than newly simulated replacements.

7. Bonus experiments: bidirectionality and attention

The assignment bonus asks for experiments with GRU architectures, bidirectional recurrent networks and temporal attention pooling. The supplied notebook evaluates five bonus configurations and compares them with the plain recurrent baselines.

Bonus configuration results

Configuration	What changes	Validation %	Test %
BiLSTM-L1-H128	Bidirectional LSTM	95.83	96.95
BiGRU-L1-H128	Bidirectional GRU	98.50	96.73
LSTM-L1-H128+Att	LSTM with temporal attention	96.33	95.32
GRU-L1-H128+Att	GRU with temporal attention	97.33	96.68
BiGRU-L2-H128+Att	2-layer bidirectional GRU + attention	99.33	97.95

Bidirectional processing. The bidirectional LSTM reached 96.95% test accuracy and the bidirectional GRU reached 96.73%. Processing the sequence in both directions provides context from earlier and later frames, which can help distinguish acoustically similar digits.

Temporal attention. Attention learns weights over valid timestep representations and forms a weighted summary of the utterance. The attention-based GRU reached 96.68%, compared with 94.82% for the plain one-layer GRU with 128 units. The attention-based LSTM reached 95.32%, compared with 94.27% for the corresponding plain LSTM.

Combined bonus model. BiGRU-L2-H128+Att achieved the best result: 99.33% validation accuracy and 97.95% test accuracy. This is an absolute gain of 2.68 percentage points over the best plain GRU (95.27%). The combined model has 474,890 parameters, so the gain may reflect both the architecture and its greater capacity.

Bonus-task takeaway. The experiments support combining bidirectional context with temporal attention for this dataset. However, each configuration was trained once with one random seed; repeated runs would be needed to determine whether small differences between bonus models are stable.

Bonus Experiment: Temporal Attention Visualization

The reference report's bonus experiment visualizes temporal attention weights for one example from each of the ten digit classes. The top row shows the digit samples, while the orange bars below indicate the attention assigned to each input row by the GRU-L1-H128+Att model.

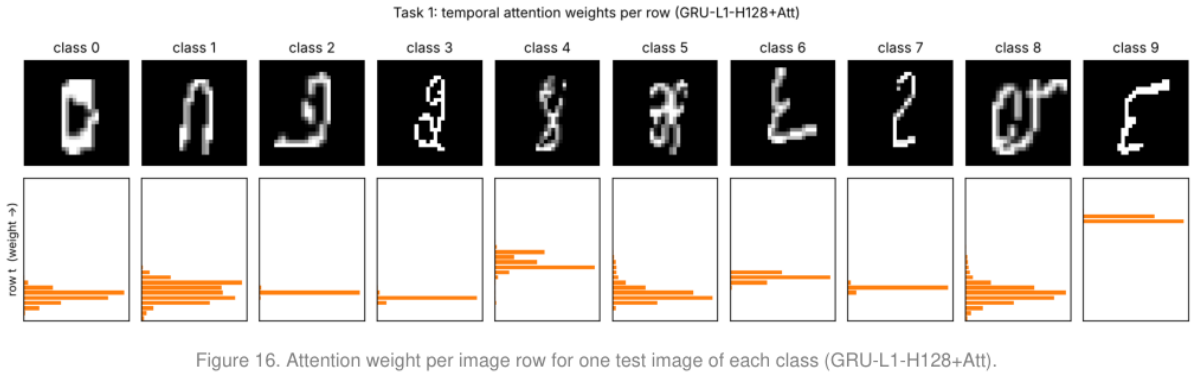


Figure 16. Attention weight per image row for one test image of each class (GRU-L1-H128+Att).

Figure 16. Temporal attention weights per image row for one test image of each class (GRU-L1-H128+Att), reproduced from the supplied reference PDF.

Interpretation. Attention is concentrated on selected rows rather than distributed uniformly. The pattern differs across classes, suggesting that the model assigns different importance to different portions of the input sequence. This visualization is reproduced from the reference report and is a Kannada-MNIST Task 1 figure; it is included here to preserve the requested same-class bonus output.